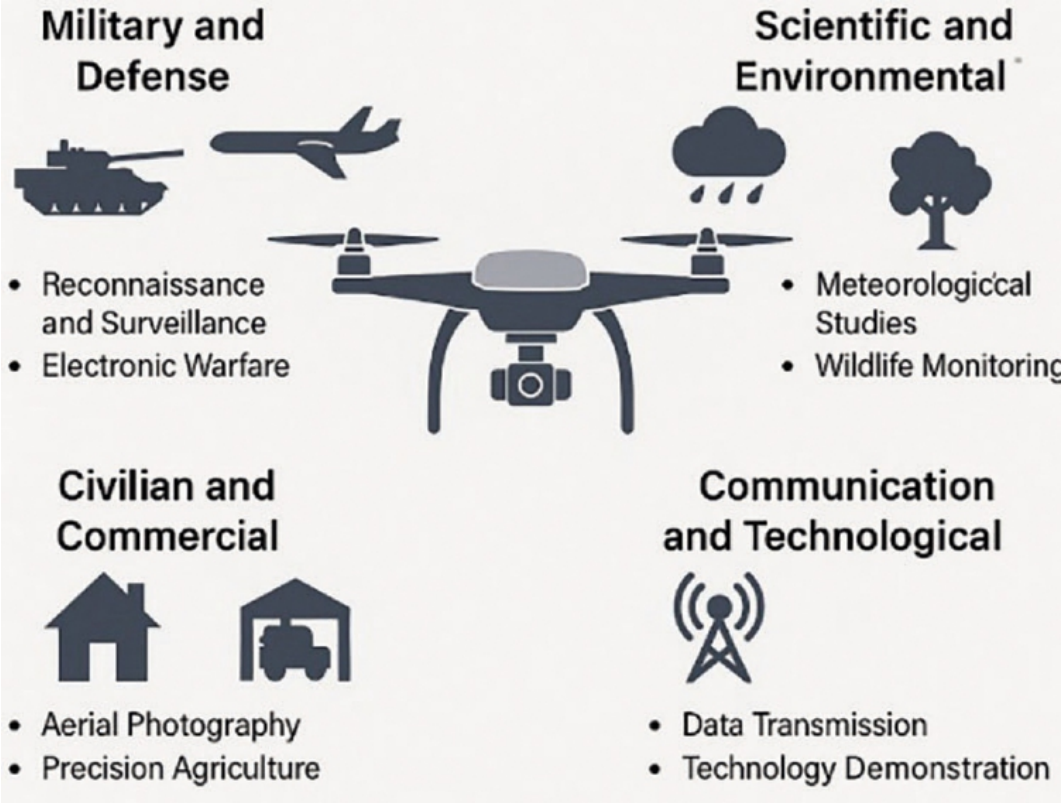


Figure 3.3 Applications of second-generation UAV systems.

3.5 Developments and Pioneers of Second-generation Drone Models

The development of second-generation drone models marked a period of extraordinary technological progress, driven by advancements in aeronautics, electronics, and computing, as well as the growing strategic need for reliable, pilotless aerial systems. Spanning from the 1950s through the early 2000s, this generation witnessed the transformation of unmanned aerial vehicles (UAVs) from basic radio-controlled aircraft to sophisticated reconnaissance and surveillance platforms capable of long-range, semi-autonomous operations. This evolution was shaped by visionary engineers, defense organizations, and aerospace industries that laid the groundwork for the modern UAV systems used today.